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Aggravating effects of alcohol outlet types on street robbery and aggravated assault in New York City

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ABSTRACT

The present study uses Risk Terrain Modeling (RTM) to analyze how nine different alcohol outlet types differentially influence the likelihood of aggravated assault and street robbery at micro-level places in New York City. Separate models were conducted for each of New York's five boroughs to account for the differing environmental characteristics across the city. Results suggest that spatial influence of crime risk differed across alcohol outlet types, with off-premise grocery stores having the largest influence on the likelihood of both aggravated assault and street robbery and on-premise bars/taverns having a strong influence only on the likelihood of street robbery. In addition, although certain alcohol outlets affected crime in each borough, the spatial influence of others was more volatile. It is clear not all alcohol outlet types are equally criminogenic across boroughs. Thus, crime reduction efforts should be unique and localized according to the risk associated with outlet types and place.

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Alcohol outlets; Risk Terrain Modeling; environmental criminology

Introduction

The alcohol–violence link is well established in social science research, with a strong emphasis on the geospatial influence of alcohol outlets on violence. Existing research has found that higher levels of alcohol outlet density are significantly associated with higher rates of violence and street crimes (Gorman et al. 2018; Gruenewald and Remer 2006; Livingston 2008a, 2008b; Norstrõm, 2000; Pridemore and Grubesic 2013; Roman et al. 2008; White, Gainey, and Triplett 2015; Zhu, Gorman, and Horel, 2004). Alcohol outlets have also been geographically linked with interpersonal and domestic violence (Livingston 2011b; Cunradi et al. 2012; Waller et al. 2012; Snowden 2016), suicide mortality (Värnik et al. 2007), and disadvantaged neighborhood characteristics (Alaniz, Cartmill, and Parker 1998; Sampson and Raudenbush 1999; Foster et al. 2017; Nielsen et al. 2010).

Existing research has begun to examine the spatial influence of different types of alcohol outlets on crime and have found evidence to suggest alcohol outlets are more nuanced, and that not all types are equally criminogenic (Gmel, Holmes, and Studer 2016). However, previous studies are limited in their method of disaggregation. Research that has disaggregated overall alcohol outlets has primarily compared on-premise and off-premise facilities (Speer et al. 1998; Costanza, Bankston, and Shihadeh 2001; Livingston 2008a, 2008b, 2011a, 2011b; Branas et al. 2009; Snowden 2016). On-premise facilities are generally categorized as alcohol outlets that are permitted the sale of alcohol, but the sales are limited to patron consumption on the premise of the facility, as opposed to off-premise facilities that

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are generally categorized as alcohol outlets permitted the sale and distribution of alcohol off the premise of the facility.

As argued by Gmel, Holmes, and Studer (2016), research examining the relationship between alcohol outlets and violence should examine in detail different outlet types to capture the full range of individual differences in alcohol outlets and to develop specialized policies aimed at specific outlet types. In this vein, more recent studies have compared the effect of traditional types of on-premise alcohol outlets, such as bars or restaurants, but neglect the difference within different types of off-premise liquor outlets (Scribner, MacKinnon, and Dwyer 1994, 1995; Lipton and Gruenewald 2002; Gruenewald and Remer 2006; Gruenewald et al. 2006; Cunradi et al. 2012; Pridemore and Grubesic 2013; Snowden and Pridemore 2013; Cameron et al. 2016).

The current study contributes to the literature through a comparative analysis of the nine separate types of licensed alcohol outlets in New York City (NYC). The disaggregated alcohol outlet types included in our study moves beyond the conventional typologies that have typically been used in prior research. We believe this approach offers insights into unique spatial influence exerted by different alcohol outlets and the associated policy implications for crime prevention practice. Using Risk Terrain Modeling (RTM), we tested the effect of each establishment type on aggravated assault and street robbery. Separate models were conducted for each of the five boroughs of NYC to test whether the effect of alcohol outlets is at least somewhat contingent on the larger social context in which they exist. Our findings suggest that not all alcohol outlets are equally criminogenic and must be examined separately to identify important nuances in each type of outlet. Furthermore, the spatial influence of alcohol outlet types on crime is somewhat different from borough to borough, suggesting the social context of alcohol outlets still matters. Discussion of the how alcohol outlets differ and implications for crime prevention are also presented.

Review of relevant literature

Research on the geographic distribution of crime has emphasized the analysis of place-based features of the environment and their relation to crime occurrence. This body of research adheres to the Environmental Criminology framework, a family of theories concerned with the immediate circumstances in which crime occurs (Wortley and Mazerolle 2008). Routine Activities Theory (Cohen and Felson 1979) considers crime as the outcome of the spatial and temporal convergence of a likely offender and a suitable target in the absence of a capable guardian. The convergence of these three elements is largely determined by behavioral patterns of the population. Crime Pattern Theory explains behavioral patterns as influenced by three types of activity spaces: nodes (places where people spend most of their time such as home, work, and places of recreation), paths (travel routes between nodes), and edges (boundaries between different areas) (Brantingham and Brantingham, 1995). Across the landscape, these activity spaces collectively comprise the environmental backcloth, which Brantingham and Brantingham (1993) define as the environmental features that comprise a specific place and exert influence on human behavior occurring within said place. Activity spaces can be made criminogenic by the presence of crime attractors and crime generators. Crime attractors are places inherently criminogenic and affording many opportunities for deviance that are well known to offenders (Brantingham and Brantingham, 1995; Roman et al. 2008; Bernasco and Block 2009). Conversely, crime generators function as places that are not inherently criminogenic, but rather attract large numbers of people for reasons unrelated to criminal motivations (Clarke and Eck 2005).

The routine activity and crime pattern theories suggest particular casual mechanisms by which alcohol outlets can contribute to crime. As per routine activities, different types of alcohol outlets may offer different levels of capable guardians based on expected levels of deviant behavior. Alcohol establishments with less security may serve as likely places where motivated offenders and potential victims converge, and where conventional norms are suppressed in favor of morally ambiguous or deviant behavior (Madensen and Eck 2008; Roncek and Maier 1991). In addition, alcohol outlets can function as both crime generators and crime attractors depending upon their unique characteristics. Bars and nightclubs with little security, frequented by local deviants, or situated in disadvantaged neighborhoods with little community intervention may serve as hosting places for motivated offenders and potential victims to participate in other criminal activity such as drug dealing or prostitution (Alaniz, Cartmill, and Parker 1998; Pridemore and Grubesic 2013). These decisions, along with themes, locations, and marketing strategies, are collectively known as place-manager decisions that have been found to create environments that may suppress or facilitate violence, and helps explain variation in violence even within the same types of alcohol outlets (Madensen and Eck 2008). Importantly, not all environmental features are associated with the same risk, just like not all types of alcohol outlets provide the same degree of risk to the environmental landscape (Caplan and Kennedy 2016). There are significant differences in the amount of risk associated with on-premise bars located in smaller communities frequented by locals and off-premise alcohol retail stores in affluent communities (Pridemore and Grubesic 2012; Snowden and Pridemore 2013; Cameron et al. 2016). Similarly, there may be differences in the amount of risk between these outlets and late-night dance clubs located in densely populated areas.

In consideration of such findings, not all alcohol outlet types should be expected to be equally criminogenic. Alcohol outlets vary across a range of variables that may be related to crime, including location, management, and alcohol consumption, to name a few. For example, some on-site establishments, such as bars and taverns, are designed where the primary activity and main attraction is alcohol consumption, whereas other on-site establishments, such as restaurants or cafés, are designed to focus on food or the dining experience rather than emphasizing alcohol consumption (Snowden and Pridemore 2013). These differences matter because they challenge the assumption that alcohol availability, on its own, generates crime. Rather, this research suggests that the social context in which alcohol is present may largely determine if a particular alcohol establishment exerts a criminogenic effect on the surrounding environment. Thus, the social relevancy (Caplan and Kennedy 2016; Irvin-Erickson 2014) of different types of alcohol establishments may influence nearby human behaviors differently, and therefore affect the nuanced spatial patterns of related crime events.

Prior research typically neglects this difference by aggregating all types of alcohol outlets in their analysis. However, a growing number of studies distinguish between different typologies of alcohol establishments. Several studies have compared the criminogenic effect of on-premise and off-premise alcohol outlets (Gorman, Speer, Labouvie, and Subaiya, 1998; Costanza, Bankston, and Shihadeh 2001; Lipton and Gruenewald 2002; Gruenewald and Remer 2006; Livingston 2011a). A study by Scribner et al. (1999) found that both off-premise and on-premise alcohol outlet densities in New Orleans were significantly related to total homicides in 1994–1995, but off-premise outlet densities had a higher effect size ($\beta = .58$) relative to on-premise outlet densities ($\beta = .27$). In fact, a 10% higher density of off-premise outlets accounted for a 2.4% higher homicide rate in associated census tracts (Scribner et al. 1999). High density of off-premise alcohol outlets and bars have consistently been linked with higher rates of violence across local areas, with each additional off-premise or bar establishment increasing the rate of reported violent event by up to 2–3% (Gruenewald and Remer 2006; Cameron et al. 2016), whereas restaurants and cafés have a negative relationship with local rate of violence (Gruenewald and Remer 2006). Branas, Elliot, Richmond, Culhane, and Wiebe (2009) found that being in an area of high off-premise alcohol outlets increased the risk of being shot by 2 times, whereas areas with high on-premise alcohol outlets had no effect on risk of being assaulted with a gun. Different alcohol outlet types have also been found to have varying effects on the seriousness of assault. Snowden and Pridemore (2013) found that high spatial densities of on-premise alcohol outlets including restaurants and bars were positively associated with high density areas of simple assault, whereas off-premise liquor retail establishments were positively associated with high density areas of aggravated assault. Pridemore and Grubesic (2013) found that the alcohol outlet-violence association was strongest when examining off-premise outlets relative to on-premise, and for simple assault relative to aggravated assault. Recently, Gorman et al. (2018) found that on-premise alcohol outlets had a small but significant positive association with counts of violent crime at the block group level, and off-premise alcohol outlets had a similar association with rates of violence in adjacent areas.

In their critical review, Gmel, Holmes, and Studer (2016) found research has mostly considered outlet types within highly aggregated typologies (e.g., off-premise outlets and on-premise outlets), thus overlooking the wide diversity of different outlet types. Scholars have argued that off-premise alcohol outlets (e.g., liquor, convenience, and grocery stores) operate similarly as potential gathering places of social disorder (Pridemore and Grubesic 2012). The present study builds on this, arguing there are fundamental differences in the behavior of patrons and physical designs between similar off-premise alcohol outlet. For example, there are differences in the management and training of staff in handling alcohol sales between an off-premise grocery store with a liquor license, where alcohol availability is typically limited to beer and wine, and an off-premise alcohol retail store, where harder liquor spirits are readily available. Recent research has also begun disaggregating on-premise alcohol outlets (e.g., restaurants and bars/taverns), but these efforts could be taken further (Scribner, MacKinnon, Dwyer, 1994, 1995; Gruenewald and Remer 2006; Cunradi et al. 2012; Snowden and Pridemore 2013; Cameron et al. 2016). There are differences in what constitutes as 'normative' behavior of patrons between on-premise bars/taverns, where most patrons may leisurely come together over drinks in a wide space, and on-premise nightclubs, where most patrons may congregate over cocktails and dance on crowded dance floors. Physical design is also different depending on the type of alcohol outlet. Most nightclubs are meant to hold large-capacity of patrons in loud, dimly lit, often overcrowded, and high temperature structures. These conditions of physical design in nightclubs have been linked to high levels of aggression (Macintyre and Homel 1997). In contrast, most bars and taverns are designed to accommodate patrons with adequate seating and space.

Researchers have also measured how neighborhood characteristics influence the criminogenic nature of alcohol outlets, oftentimes generating mixed results (Gorman et al. 2001; Gruenwald et al., 2006; Pridemore and Grubesic 2012, 2013). However, although the effect of community context on the spatial influence of alcohol outlets has been extensively examined, less is known about how community characteristics impact the spatial influence of such a disaggregate level of alcohol outlet types as the present study. Future research is needed for different types of alcohol outlets to properly capture the nuance of how alcohol outlet types influence the likelihood of crime. Barnum et al. (2017) argues each area has a unique environmental backcloth, and the ways in which certain place features such as alcohol outlets come together to influence criminal opportunities is different across jurisdiction based on the interaction of many social or ecological confounding variables. New York City, the setting for the present study, is an example of a unique environment, comprising of five separate boroughs that vary across social, economic, and ecological factors. Examining the spatial influence of alcohol outlets across each borough separately captures the unique interaction of how different outlet types are associated with crime across different neighborhood environments.

The current study

The present study contributes to the literature by examining the spatial influence of different types of alcohol outlets on the occurrence of aggravated assault and street robberies in the five boroughs of NYC during the one-year study period from 10/01/2014 to 09/30/2015. The analysis incorporates RTM to measure and compare the spatial influence of different types of alcohol outlets in NYC. A total of nine different alcohol outlet types were included in our spatial analysis, moving beyond the conventional typologies that have typically been used in prior research (e.g., off-premise vs. on-premise, bars vs. restaurants, etc.). To the best of our knowledge, no previous study has included a disaggregate analysis of alcohol outlets as detailed as our present study. Such a detailed disaggregate analysis offers insight into unique policy implications for each significant alcohol outlet type.

Similar to Haberman, Groff, and Taylor's (2013) analysis of public housing communities, we argue that each alcohol outlet possesses different characteristics that moderate criminogenic opportunities inherent to alcohol outlets, and simply comparing on on- vs. off-premise alcohol outlets is not enough. Other alcohol outlets such as drug stores (off-premise) and hotels (on-premise) have historically been ignored or condensed to fit the conventional alcohol outlet typology. The present study hypothesizes that different alcohol outlet types will have varying effects on different crime types across different boroughs.

Study setting

New York City is the largest metropolitan city in the United States, with a total population of over 8.5 million residents (U.S. Census Bureau, 2015) and comprising of five different boroughs: Manhattan, the Bronx, Brooklyn, Queens, and Staten Island. Each borough comprises different neighborhood characteristics that make the aggregate of NYC diverse in social, economic, educational, and demographic ways. This also suggests that different boroughs may have different levels of disaggregated alcohol outlets within them, and thus different social relevancies related to attractive illegal behavior and crime. Data on county-level neighborhood characteristics were collected from the US Census Bureau's American Community Survey 5-year estimates (2011–2015). We present five neighborhood characteristics to highlight the diversity of each borough in NYC (see Table 1a): percentage of persons living below the poverty line, population density, median household income, unemployment rate, and an ethnic heterogeneity index.[1]

As environments change, so too do the social and ecological conditions optimal for crime opportunities. Thus, it is important to examine the influence of place features across different settings to investigate how crime emerge and persist under different conditions (Barnum et al. 2017). During the study period, Manhattan had the highest population density, the Bronx had more than twice the poverty rate and half the median income than Staten Island or Queens, and Brooklyn and Manhattan had more than twice the ethnic diversity as the Bronx. This diversity between the different boroughs makes NYC a unique location to examine how alcohol outlets influence crime across different environments at the same time in a standardized way. The present study runs separate models by borough as different study settings, in recognition of the possibility

Table 1. Descriptive statistics.

Category	Manhattan	Bronx	Brooklyn	Queens	Staten Island
(a) Neighborhood Characteristics					
Population density	71,375.68	33,545.25	37,341.85	21,066.91	8,111.26
% Poverty	17.9%	30.7%	23.2%	15.1%	12.5%
Median income	$72,871	$34,299	$48,201	$57,720	$73,197
Unemployment rate	7.5%	14%	10%	8.6%	6.9%
Ethnic Heterogeneity	0.4983	0.1846	0.4592	0.3861	0.4671
(b) Reported Crime					
Street Robbery	N = 914	N = 1,630	N = 2,179	N = 1,370	N = 153
Aggravated Assault	N = 4,492	N = 8,552	N = 10,079	N = 6,098	N = 1,338
(c) Alcohol Outlets					
Alcohol Retail Stores (Off-Premise)	N = 351	N = 179	N = 411	N = 320	N = 57
Drug Stores (Off-Premise)	N = 224	N = 69	N = 135	N = 148	N = 38
Grocery Stores (Off-Premise)	N = 1,112	N = 1,216	N = 2,239	N = 1,669	N = 277
Bars and Taverns (On-Premise)	N = 3,392	N = 354	N = 1,398	N = 1,124	N = 233
Clubs (On-Premise)	N = 68	N = 29	N = 40	N = 40	N = 14
Eateries (On-Premise)	N = 144	N = 42	N = 122	N = 119	N = 36
Hotels (On-Premise)	N = 207	N = –	N = 11	N = 21	N = 2
Other Entities (On-Premise)	N = 29	N = 2	N = 2	N = 6	N = 1
Restaurants (On-Premise)	N = 1,322	N = 132	N = 576	N = 648	N = 77

that different types of alcohol outlets may have varying effect on reported aggravated assault and street robbery, and this spatial influence may be heterogeneous across boroughs.

Dependent variables

The previous literature on the alcohol outlet–violence association have argued that increased density of alcohol outlets within a geographic unit is significantly associated with higher rates of assaultive violence (Cameron et al. 2016; Snowden and Pridemore 2013; Pridemore and Grubesic 2013; Branas et al. 2009; Britt, Carlin, Toomey, Wagenaar, 2005). In line with previous studies, dependent variables for the present study are focused exclusively on aggravated assault and street robbery events because they best demonstrate the possible outcomes of the immediate space of liquor establishments. To better tailor the outcome measure to the analysis, we excluded all crimes occurring within residential settings from the analysis. This was because the physical space of liquor establishments is assumed to influence criminal behavior in public or quasi-public places, not within private settings. The New York City Police Department (NYPD) provided all crime records that were reported during the 10/01/2014–09/30/2015 study period. A total of 6,246 street robberies (73 robberies per 100,000 residents) and 30,559 aggravated assaults (357 aggravated assaults per 100,000 residents) were reported to the NYPD during the study period. Data were geocoded by the NYPD, with XY coordinates provided for each robbery and aggravated assault incident included in the database. The distribution of reported street robberies and aggravated assault is presented in Table 1b. As can be seen, Brooklyn had the highest reported street robbery with 2,179 cases, and Staten Island had the fewest with 153 cases. This trend continued when comparing reported cases of aggravated assault across borough, with Brooklyn accounting for the most incidents (N = 10,079) and Staten Island accounting for the least (N = 1,338).

Independent variables

The NYPD provided a list of all active alcohol outlets during the study period. The data contained the *XY* coordinates for all businesses in the database, which the research team used to map each facility within ArcGIS (version 10.3). We disaggregated the database into nine separate alcohol outlet types for use as independent variables in our analysis based on the categories identified by all liquor license types available for retail and entertainment establishments in NYC according to the New York State Liquor Authority (SLA). The disaggregated facility types are alcohol retail stores (off-premise), drug stores (off-premise), grocery stores (off-premise), bars and taverns (on-premise), night clubs (on-premise), eateries (on-premise), hotels (on-premise), other eateries (on-premise), and restaurants (on-premise). With the exception of the Bronx, which had no liquor licensed hotel establishments at the time of the study period, each borough contained all nine alcohol outlet types. All liquor establishments selected for the present study are mutually exclusive and were identified based on their main licensing type. For example, hotels with restaurants are identified as a ‘on-premise hotel’ liquor license rather than ‘on-premise restaurant’ liquor license according to the SLA license classes (SLA, 2015). These alcohol outlets are also important for policy purposes as they can influence practices not only based on outlet type, but by informing regulations at the city licensing level. Although not all alcohol outlets included in our model have been previously studied in relation to violence, all establishments included in the present analysis do carry and sell alcohol at varying levels. Further separating alcohol outlet types into these nine types is in line with arguments presented in previous studies advocating for separation of similarly structural environmental risk factors, and takes it a step further by identifying all possible incarnations of alcohol outlets (Gmel, Holmes, and Studer 2016).

The distribution of reported alcohol outlets is presented in Table 1c. As can be seen, the number of different types of liquor licensed establishments varies by borough. Off-premise grocery stores are the leaders in alcohol outlets in most boroughs. This is important given that grocery stores have been either ignored or aggregated in previous alcohol outlet-violence research. It is not surprising that off-premise alcohol retail stores, bars and taverns, and restaurants have been examined in relation to violence, as those alcohol outlets are most common in terms of alcohol availability. However, as can be seen, there are alternative establishments that present access to alcohol as well. By identifying all liquor license establishment types, the present study is able to identify the nuance between different alcohol outlets in relation to violence at a level not previously explored.

Risk terrain model

The present study applies a RTM approach to test the nearby spatial risks of different types of alcohol outlets on crime occurrence. Developed by Caplan, Kennedy, and Miller (2011), RTM is a geospatial method used to aid in crime forecasting by incorporating potential risk factors of crimes and standardizing these factors to common geographic units over a continuous surface. The technical approach to RTM begins by identifying potential risk factors (in this study, disaggregate alcohol outlets) of interest related to an outcome (i.e., crime) for which risk is being assessed (Caplan and Kennedy 2016). Each risk factor is then operationalized to a common geography by assigning a value signifying the proximity or density of each factor at every place throughout the given geography. Previous studies examining the spatial influence of alcohol outlets on crime have exclusively examined the concentration of features (i.e., spatial density), while ignoring the effect of geographic proximity (Pridemore and Grubesic 2012; Snowden and Pridemore 2013; White, Gainey, and Triplett 2015). As is evident in more recent geospatial studies, the risk of crime has been found to be higher at places as a function of proximity to alcohol outlets, at places with a dense concentration of clustering of alcohol outlets, or both depending on the conditions of the environment (Ratcliffe, 2012; Barnum et al. 2017). Furthermore, as suggested by Groff and Lockwood (2014), the spatial influence of place-based features is geographically limited with distance. Previous studies examining the spatial influence of place-based features on crime have argued that different types of features have different spatial distance (Ratcliffe, 2012). The benefit of RTM is that it accounts for spatial influence of different alcohol outlets across a range of spatial distance scale and across different spatial operationalization.

Each factor is represented by a separate coverage (raster) map of the same geography. When all map layers are combined in a Geographic Information System (GIS), they produce a composite (risk terrain) map where every place throughout the geography is assigned a composite risk value that accounts for all factors associated with the crime outcome. The higher the risk value the greater the likelihood of a crime event occurring at that location.

Model parameters

The present study focuses on the individual spatial influence of each alcohol outlet on risk of aggravated assault and street robbery in each borough of NYC. To assign context in the discussion of comparing spatial influence of place-based features on crime across different study settings, each setting must be examined under a standardized geographic metric, such as block lengths. Each borough has a different measure of block lengths, and so the different cell sizes for each model is a consequence of the different geographic conditions across jurisdictions. In conducting RTM, we used the RTMDx Utility developed by the Rutgers Center on Public Security (Caplan and Kennedy 2013). Each borough was modeled as a contiguous grid of equally sized cells about the length of half a city block, as measured within ArcGIS: Bronx, 165 ft. by 165 ft. (N = 45,080); Manhattan, 161 ft. by 161 ft. (N = 25,937); Brooklyn, 181 ft. by 181 ft. (N = 61,361); Queens, 167 ft. by 167 ft. (N = 112,281); and Staten Island, 176 ft. by 176 ft. (N = 53,457). Constructing cell size as approximately one-half of the average block allowed for analyses at the half-block increment and measuring spatial influence up to three blocks. As discussed earlier, each borough will be evaluated as its own study setting, and therefore the grid size for each model will be different because the average street length of each study setting is different. This allows a standardized comparison between the different boroughs of NYC.

Statistical analysis

RTMDx identifies the optimal spatial influence and operationalization (Caplan 2011)[2] for each risk factor as well as a Relative Risk Value (RRV). The RRV is calculated by exponentiating risk factor coefficients provided by the RTM, and can be considered as a weighting value used to compare the effect of risk factors with one another. RTMDx tests risk factor influence through a penalized regression model, with crime as the dependent variable and various operationalizations of alcohol outlet types as the independent variables. RTMDx then measures whether each raster cell is within a certain distance of different types of alcohol outlets (i.e., proximity) or in an area of high concentration of different alcohol outlet types (i.e., density). Each risk factor was tested at half-block increments out to three blocks. Through this method, the nine alcohol outlet types generated up to 108 variables for each borough that were tested for significance.[3] Incorporating 108 covariates in a single model may present problems with multiple comparisons, in that we may detect spurious correlation simply due to the number of variables tested. The penalized regression method used by RTMDx alleviates potential problems with spurious correlation due to multiple comparisons by reducing the large set of variables to a smaller set of variables with non-zero coefficients. This is accomplished through an elastic net method that forms five stratified folds from the raster cells, balancing crime counts between the folds. This balancing process is done to ensure that there is some variance across the folds to aid in the numeric stability of the modeling process.

In total, the current study consists of 10 separate RTM models, with each model representing one of the five NYC boroughs across the two crime types. RTMDx selected between 20 and 31 potentially useful variables in the robbery models and between 8 and 21 potentially useful variables in the aggravated assault models (see Table 2). These variables were then utilized in a bidirectional step-wise regression process to determine the final model type. Following a null model with no model factors, RTMDx adds each variable to the null model and re-measures the Bayesian Information Criteria (BIC) score to identify the most parsimonious combination of variables. After each iteration, the model with the lowest BIC score is selected as the new candidate model (the model to surpass). RTMDx repeats the process, adding and removing variables one step at a time, until no variable addition/removal surpasses the previous BIC score. RTMDx repeats this process with two stepwise regression models: one Poisson and one negative binomial, selecting the best model with the lowest BIC score. RTMDx also produces a relative risk value that can be interpreted as the weight of the individual risk factor, and therefore may be used for comparison across all risk factors (for more information on the statistical procedure of RTMDx, see Heffner, 2013).[4]

Table 2. Number of variables in risk terrain models.

		Street Robbery	Aggravated Assault
	# Variables	Potentially Useful	Potentially Useful
Manhattan	108	24	8
Bronx	96	31	15
Brooklyn	108	22	10
Queens	108	24	15
Staten Island	108	20	21

Results

The purpose of the present study is to identify significant spatial influence of different types of alcohol outlets across different jurisdictions. It is more important within the scope of this study to first identify which types of alcohol outlets have a significant spatial influence on aggravated assault and street robberies between boroughs than to examine how the spatial influence of different alcohol outlet types may vary based on community characteristics within boroughs. This is not to say community context has no effect on criminal behavior, but rather alcohol outlets can create settings that may suppress the effect of negative outside influences on violence, and these settings are determined by business-related decisions based on different types of establishments (Madensen and Eck 2008). Tables 3 and 4 display the results from the RTM analysis. Among the nine alcohol outlets tested, four alcohol outlets were found to be associated with aggravated assault across the different boroughs: grocery stores (off-premise), alcohol retail stores (off-premise) , eateries (on-premise), and restaurants (on-premise). Among the nine alcohol outlets tested, six alcohol outlets were found to be associated with street robberies across the different boroughs: grocery stores (off-premise), alcohol retail stores (off-premise), drug stores (off-premise), eateries (on-premise), bars and taverns (on-premise), and restaurants (on-premise).

There is clear overlap in the findings, as all four of the alcohol outlets found to be significantly associated with aggravated assault across the different boroughs were also found to be significantly associated with street robberies across the different boroughs. Interestingly, at least three alcohol outlet types included in our analyses are not spatially associated with increased likelihood of aggravated assault or street robbery in any borough: night clubs (on-premise), hotels (on-premise), and other eateries (on-premise). From the tables below, significant risk factors should be interpreted according to their operationalization and spatial influence associated for each alcohol outlet in each borough. For example, places in Queens with a high density of restaurant alcohol outlets within a radius of 0.5 block are more than 1.75 times more likely to experience aggravated assault incidences than other places. Similarly, places in Manhattan within a 2-block proximity to off-premise grocery store alcohol outlets are about 6.25 times more likely to experience street robbery incidents than other places.

Interestingly, grocery stores were found to be significantly related to both aggravated assault and street robbery at each borough, and had a larger relative risk value than any other significant alcohol outlet. Places in the Bronx within a 2-block proximity to off-premise grocery store alcohol outlets were nearly four times more likely to experience street robbery incidents than the next significant alcohol outlet type (off-premise alcohol retail stores). In Manhattan, places within a 2.5 block proximity to off-premise grocery store alcohol outlets are over 11 times more likely to experience aggravated assault incidents than other places. Queens consistently had the largest number of alcohol outlets associated with aggravated assault and street robbery. Consistent with prior research, off-premise alcohol retail stores and on-premise restaurant alcohol outlets were found to be significantly related to both aggravated assault and street robbery incidents, but this association was not found to be significant in all boroughs. On-premise bars and taverns were also found to be significantly associated with street robbery incidents, but not with aggravated assault or in the Bronx. The mixed results identified in the present study from prior research highlight the nuances in the different types of alcohol outlets and their relation to crime. Adhering to the conventional off-premise and on-premise typology would yield similar results to prior research but would miss the variation in alcohol outlet types that we observed.

Discussion

Prior research on the alcohol outlet–violence association have consistently yielded similar results: there is a positive and significant association between total outlet density and total measures of violence. In recent years, this association has been examined more closely based on alcohol outlet type. The findings in the present study appear somewhat consistent with existing research regarding off-premise versus on-premise alcohol outlet types. Our study found that both off-

Table 3. Risk factor testing for aggravated assault by borough.

	Manhattan			Bronx			Brooklyn			Queens			Staten Island		
	Op.	S.I.	RRV	Op.	S.I.	RRV	Op.	S.I.	RRV	Op.	S.I.	RRV	Op.	S.I.	RRV
Grocery Stores (Off-Premise)	Prox.	2.5 Blks.	11.31	Prox.	2 Blks.	5.48	Prox.	2 Blks.	3.40	Prox.	3 Blks.	4.23	Prox.	3 Blks.	4.81
Alcohol Retail Stores (Off-Premise)	–	–	–	Prox.	3 Blks.	1.89	Prox.	3 Blks.	1.45	Prox.	3 Blks.	1.45	–	–	–
Eateries (On-Premise)	–	–	–	–	–	–	–	–	–	Prox.	3 Blks.	1.39	Dens.	3 Blks.	1.50
Restaurants (On-Premise)	–	–	–	–	–	–	–	–	–	Dens.	0.5 Blk.	1.87	–	–	–

Abbreviations: Op., Operationalization; Dens, Density; Prox., Proximity; S.I., Spatial Influence; RRV, Relative Risk Value.

Table 4. Risk factor testing for street robbery by borough.

	Manhattan			Bronx			Brooklyn			Queens			Staten Island		
	Op.	S.I.	RRV	Op.	S.I.	RRV	Op.	S.I.	RRV	Op.	S.I.	RRV	Op.	S.I.	RRV
Grocery Stores (Off-Premise)	Prox.	2 Blks.	6.24	Prox.	2 Blks.	6.26	Prox.	2.5 Blks.	5.29	Prox.	2.5 Blks.	4.72	Dens.	3 Blks.	9.28
Alcohol Retail Stores (Off-Premise)	–	–	–	Dens.	1.5 Blks.	1.69	Prox.	3 Blks.	1.50	Prox.	3 Blks.	1.65	–	–	–
Drug Stores (Off-Premise)	Dens.	0.5 Blk.	2.89	–	–	–	Dens.	1 Blk.	1.64	Prox.	1 Blk.	1.67	–	–	–
Bars and Taverns (On-Premise)	Dens	0.5 Blk.	1.88	–	–	–	Dens.	0.5 Blk.	2.27	Dens.	0.5 Blk.	4.68	Prox.	0.5 Blk.	7.29
Eateries (On-Premise)	–	–	–	Prox.	3 Blks.	1.55	–	–	–	Prox.	3 Blks.	1.83	–	–	–
Restaurants (On-Premise)	Dens.	0.5 Blk.	1.67	Prox.	3 Blks.	1.54	Dens.	0.5 Blk.	1.90	Prox.	0.5 Blk.	1.49	–	–	–

Abbreviations: Op., Operationalization; Dens, Density; Prox., Proximity; S.I., Spatial Influence; RRV, Relative Risk Value.

premise alcohol retail stores and on-premise restaurant alcohol outlets are significantly associated with street robbery and aggravated assault; however, on-premise bars/taverns are not significantly associated with aggravated assault (Gruenewald et al. 2006; Cameron et al., 2016; Snowden and Pridemore 2013). This may be due to the presence of a capable guardian in the form of bouncers or security guards at most bars and taverns, but mostly absent from restaurants. This highlights an important distinction within on-site establishments: some on-site establishments, such as bars and taverns, are designed where the primary activity and main attraction is alcohol consumption, whereas other on-site establishments, such as restaurants or cafés, are designed to focus on food or the dining experience (Snowden and Pridemore 2013). As a result, there is a reasonable expectation for greater security presence at bars and taverns to watch over patrons and greater alcohol consumption, compared to restaurants.

Results further show off-premise alcohol outlets are consistently associated with higher risk of aggravated assault and street robbery across all boroughs, whereas on-premise alcohol outlets have different effects on the risk of crime across different boroughs. Particularly, off-premise grocery stores were consistently found to be the strongest predictor of aggravated assault and street robbery in all five NYC boroughs. Off-premise alcohol retail stores are also found to be a strong alcohol outlet predictor of aggravated assault and street robbery in the Bronx, Brooklyn, and Queens. On-premise eateries and on-premise restaurants are also found to be strong alcohol outlet predictors of both aggravated assault and street robbery, but this association seems to be mediated by some form of social context because the same results are not present for every borough. On-premise alcohol outlets also appear to be stronger predictors for risk of street robbery than aggravated assault. Although on-premise restaurants are found to be a strong predictor of both aggravated assault and street robbery, this association seems to be present in more boroughs predicting risk for street robbery (i.e., strong predictor in four boroughs) than for aggravated assault (i.e., strong predictor in 1 borough). On-premise bars and taverns were found to be strong alcohol outlet predictors of street robbery in every NYC borough except the Bronx but were not found to be significantly associated with aggravated assault.

Results also show there are variations in the alcohol outlet-violence association by borough, suggesting unique neighborhood characteristics in each borough may contribute to how criminogenic different alcohol outlets may be. Both off-premise and on-premise alcohol outlets are strong predictors of aggravated assault and street robbery in the NYC borough of Queens, whereas mostly off-premise alcohol outlets are strong predictors of aggravated assault in the other four boroughs. Additionally, alcohol outlets appear to be least predictive of aggravated assault and street robbery in Staten Island relative to the other four boroughs. This makes sense when considering how the fundamental differences in daily behavior patterns across different boroughs afford different criminogenic opportunities.

Conclusion

Results from the present study suggests alcohol outlets provide opportunities for risk of crime, but the relative risk is unique to each type of outlet, the type of crime (i.e., aggravated assault or street robbery), and under certain neighborhood conditions as suggested by varying spatial influence based on the borough. As an example, Figure 1 presents the relative risk of aggravated assault and street robbery respectively based on the spatial influence of significant alcohol outlet types in Queens and Brooklyn.

Different alcohol outlets provide different challenges not only on the influence of crime, but also how to address crime problems. The clustering of the spatial influence of positively related alcohol outlet types on the risk of crime presents implications for crime-prevention strategies. Focusing prevention efforts in areas with high risk of increased likelihood of crime would be a more efficient use of police resources. Crime-prevention efforts should also be highly specialized according to the

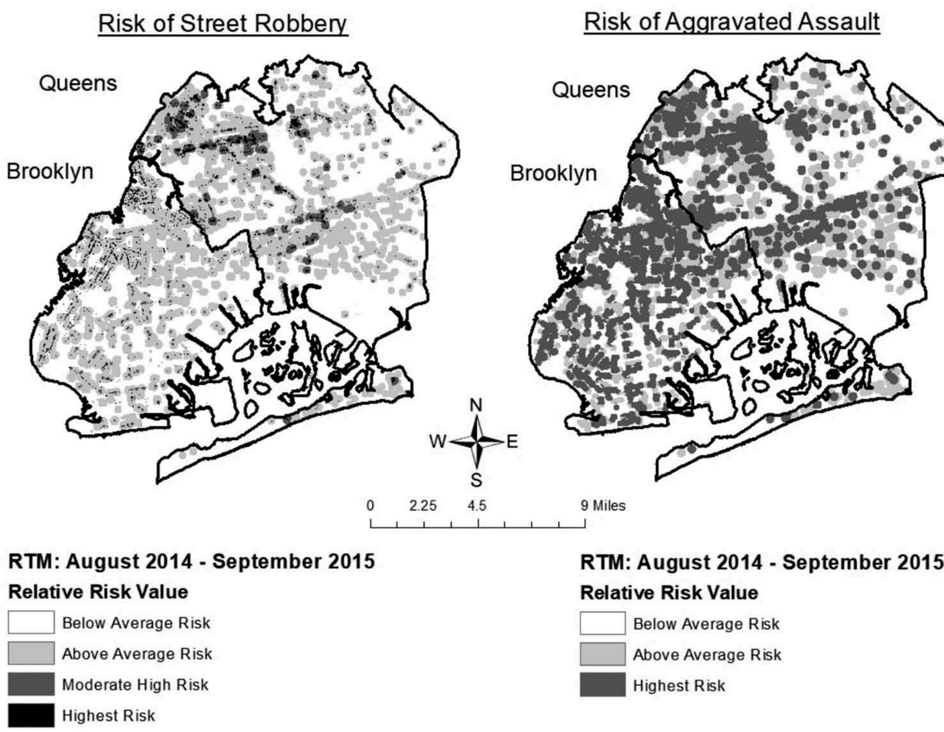

Figure 1. Relative risk of crime based on significant alcohol outlet types.

type of crime and vulnerable environmental features. According to the situational prevention literature, crime-reduction efforts should comprise of

> measures directed at highly specific forms of crime that involves management, design, or manipulation of the immediate environment in as systematic and permanent a way as possible to reduce the opportunities for crime and increase its risks as perceived by a wide range of offenders (Clarke 1983, p. 225, 1995).

This presents policy implications for crime-reduction techniques specific to each type of outlet. For example, off-premise and on-premise alcohol outlets have different expected levels of deviant behavior given fundamental differences in alcohol availability and main attraction. There is a higher level of expectation for deviant behavior in on-premise bars and taverns because alcohol is the main attraction for patrons at these establishments. Consequently, bars and taverns employ higher levels of guardianship in the form of bouncers and training staff members how to handle deviant behavior before escalation, which may limit the risk of violent encounters (e.g., aggravated assault). However, street robberies can extend beyond the physical space of the bar, limiting the crime-reduction influence of staff. Following the situational crime prevention model, additional efforts can be made through increasing the risk such as increase law enforcement presence near bars or work with bar managers to set up car service for patrons. Different from on-premise outlets, where alcohol is purchased and consumed under the supervision of staff and other security, alcohol purchased at off-premise outlets are largely consumed elsewhere without such guardianship.

It is important to highlight the practicality of identifying the spatial influence of alcohol outlets on crime from a policy framework. Neighborhood characteristics, such as poverty, employment, ethnic heterogeneity, and residential mobility, are all variables used to measure the collective efficacy and social disorganization of an environment. It is clear that these factors are consistently found to be associated with crime (Pridemore 2011; Shaw and McKay 1942). However, such factors are difficult to remedy from a policy perspective or other social mechanism. On the other hand, structuring policies around the placement of different alcohol outlet licenses, such as setting density thresholds in an area or prohibiting new establishments within a certain distance of an existing one, has been found to be more amenable (Livingston, Chikritzhs, and Room 2007; Pridemore and Grubesic 2013). In doing so, such policies can promote more responsible alcohol service and availability, while also transform the structure of a neighborhood by manipulating the placement of different alcohol outlets in a way that will reduce violence and is safer for the community.

The results presented in this study reinforce the appeal of place-based policing practices, which have consistently demonstrated effectiveness by addressing 'risky facilities' in high-crime areas (Kennedy et al. 2016). In the context of alcohol outlets, the present study argues that not all alcohol outlet provides equal criminogenic opportunities for aggravated assault or street robbery, and crime reduction efforts aimed at minimizing the effect of alcohol outlets must be tailored based on the type of alcohol outlet and the local area. The procedure of using RTM to identify target areas for place-based interventions has been found to be an effective tool (Kenndey, Caplan, and Piza, 2018). The use of RTM in such a manner requires unique partnerships with local police departments to think creatively about the design and proper implementation of these interventions (Caplan 2011). As such, the results from the present study are relevant to academics and practitioners.

Despite these implications, the current study suffers from specific limitations that should be mentioned. First, similar to Pridemore and Grubesic (2013), we have no way of knowing which crimes included in our models are alcohol-related or simply occurred at an alcohol outlet. Information about alcohol involvement is sometimes recorded as part of the police record, but is not available for use in the present study. Second, the present study does not include statistical testing of social disorganization/neighborhood characteristics. Rather, the present study ran geospatial analyses in the five different NYC boroughs, arguing different locations inherently have varying neighborhood characteristics that would yield mixed results. However, given the size of each borough, the levels of social disorganization are obviously heterogeneous across the landscape. Although incorporating social disorganization measures within the RTM was beyond the scope of the current study, future research should seek to more readily incorporate both environmental criminology and social disorganization measures in such analyses (see, for example, Piza et al. 2017). Finally, the present study does not include direct measures of alcohol availability (i.e., actual alcohol sales) or alcohol consumption that would reinforce the argument of different mechanisms in each type of alcohol outlet. Actuarial measures of alcohol availability and alcohol consumption such as sale records or establishment details (i.e., the number of workers, availability of security) is needed to appropriately operationalize the alcohol outlet–violence association. Previous studies have relied heavily on the geospatial association of outlet density, but less is known about how the prevalence of alcohol at these locations influences risk of crime. Furthermore, future research on alcohol outlets and crime should employ longitudinal analysis. Although cross-sectional studies such as ours are informative, it is important to understand how changes in outlet density influence changes in local violence rates. Nevertheless, the present study builds on previous research by identifying distinct differences in the association between types of alcohol outlets and crime, expanding beyond the off-premise/on-premise typology.

Notes

1. In line with previous studies focusing on community characteristics, the ethnic heterogeneity index was calculated as '1-(Σp_i^2), where p_i is the proportion of individuals of a given ethnic group, which is squared and summed across the groups that are distinguished (e.g., Whites and non-Whites). This index reflects the probability that two randomly drawn individuals would differ in ethnicity' (Osgood and Chambers, 2000, 93).
2. The notion of spatial influence operationalization is key, as prior research suggests that different criminogenic features exert different types of influence on crime. Research in Philadelphia, for example, found that violence is highly clustered within 85 ft of bars then dissipates rapidly (Ratcliffe, 2012), while the effect of schools, halfway houses, and drug treatment centers varies substantially by distance and crime type (Groff and Lockwood 2014). RTMDx identifies the precise distance at which the effect of the spatial risk factor on crime is maximized.

3. The exception was the Bronx model, which only had eight risk factors due to the absence of hotel with liquor licenses in the borough. The Bronx model generated 96 variables for testing.
4. Spatial autocorrelation may be problematic because significance values may be biased and the likelihood of Type I error becomes more likely. However, the RTMDx diagnostic tool employs a cross-validation technique, which deemphasizes significance testing for variable selection (Heffner, 2013: 38).

Disclosure statement

No potential conflict of interest was reported by the authors.

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References

Alaniz, M. L., R. S. Cartmill, and R. N. Parker. 1998. "Immigrants and Violence: The Importance of Neighborhood Context." *Hispanic Journal of Behavioral Sciences* 20 (2): 155–174. doi:10.1177/07399863980202002.

Barnum, J. D., J. M. Caplan, L. W. Kennedy, and E. L. Piza. 2017. "The Crime Kaleidoscope: A Cross-Jurisdictional Analysis of Place Features and Crime in Three Urban Environments." *Applied Geography* 79: 203–211. doi:10.1016/j.apgeog.2016.12.011.

Bernasco, W., and R. Block. 2009. "Where Offenders Choose to Attack: A Discrete Choice Model of Robberies in Chicago." *Criminology* 47 (1): 93–130. doi:10.1111/crim.2009.47.issue-1.

Branas, C. C., M. R. Elliott, T. S. Richmond, D. P. Culhane, and D. J. Wiebe. 2009. "Alcohol Consumption, Alcohol Outlets, and the Risk of Being Assaulted with a Gun." *Alcoholism: Clinical and Experimental Research* 33 (5): 906–915. doi:10.1111/acer.2009.33.issue-5.

Brantingham, P., and P. Brantingham. 1993. "Nodes, Paths and Edges: Considerations on The Complexity Of Crime and The Physical Environment." *Journal Of Environmental Psychology* 13 (1): 3–28. doi:10.1016/S0272-4944(05)80212-9.

Brantingham, P., and P. Brantingham. 1995. "Criminality Of Place." *European Journal on Criminal Policy and Research* 3 (3): 5–26. doi:10.1007/BF02242925.

Britt, H. R., B. P. Carlin, T. L. Toomey, and A. C. Wagenaar. 2005. "Neighborhood Level Spatial Analysis of the Relationship between Alcohol Outlet Density and Criminal Violence." *Environmental and Ecological Statistics* 12 (4): 411–426. doi:10.1007/s10651-005-1518-3.

Cameron, M. P., W. Cochrane, C. Gordon, and M. Livingston. 2016. "Alcohol Outlet Density and Violence: A Geographically Weighted Regression Approach." *Drug and Alcohol Review* 35 (3): 280–288. doi:10.1111/dar.12295.

Caplan, J. 2011. "Mapping the Spatial Influence of Crime Correlates: A Comparison of Operationalization Schemes and Implications for Crime Analysis and Criminal Justice Practice." *Cityscape* 13 (3): 57–83.

Caplan, J., and L. Kennedy. 2013. *Risk Terrain Modeling Diagnostics Utility (Version 1.0)*. Newark, NJ: Rutgers Center on Public Security. http://rutgerscps.weebly.com/software.html.

Caplan, J. M., and L. W. Kennedy. 2016. *Risk Terrain Modeling: Crime Prediction and Risk Reduction*. Oakland, CA: University of California Press.

Caplan, J. M., L. W. Kennedy, and J. Miller. 2011. "Risk Terrain Modeling: Brokering Criminological Theory and GIS Methods for Crime Forecasting." *Justice Quarterly* 28 (2): 360–381. doi:10.1080/07418825.2010.486037.

Clarke, R. V. 1983. "Situational Crime Prevention: Its Theoretical Basis and Practical Scope." *Crime and Justice* 4: 225–256. doi:10.1086/449090.

Clarke, R. V. 1995. "Situational Crime Prevention." *Crime and Justice* 19: 91–150. doi:10.1086/449230.

Clarke, R. V., and J. E. Eck. 2005. *Crime Analysis for Problem Solvers*. Washington, DC: Center for Problem Oriented Policing.

Cohen, L. E., and M. Felson. 1979. "Social Change and Crime Rate Trends: A Routine Activity Approach." *American Sociological Review* 44 (4): 588–608. doi:10.2307/2094589.

Costanza, S. E., W. B. Bankston, and E. Shihadeh. 2001. "Alcohol Availability and Violent Crime Rates: A Spatial Analysis." *Journal of Crime and Justice* 24 (1): 71–83. doi:10.1080/0735648X.2001.9721617.

Cunradi, C. B., C. Mair, W. Ponicki, and L. Remer. 2012. "Alcohol Outlet Density and Intimate Partner Violence-Related Emergency Department Visits." *Alcoholism: Clinical and Experimental Research* 36 (5): 847–853. doi:10.1111/acer.2012.36.issue-5.

Foster, S., P. Hooper, M. Knuiman, G. Trapp, and L. Wood. 2017. "Does Alcohol Outlet Density Differ by Area-Level Disadvantage in Metropolitan Perth?" *Drug and Alcohol Review*. doi:10.1111/dar.12469.

Gmel, G., J. Holmes, and J. Studer. 2016. "Are Alcohol Outlet Densities Strongly Associated with Alcohol-Related Outcomes? A Critical Review of Recent Evidence." *Drug and Alcohol Review* 35 (1): 40–45. doi:10.1111/dar.12304.

Gorman, D. M., P. W. Speer, E. W. Labouvie, and A. P. Subaiya. 1998. "Risk Of Assaultive Violence and Alcohol Availability in New Jersey." *American Journal Of Public Health* 88 (1): 97-100. doi:10.2105/AJPH.88.1.97.

Gorman, D. M., P. W. Speer, P. J. Gruenewald, and E. W. Labouvie. 2001. "Spatial Dynamics of Alcohol Availability, Neighborhood Structure and Violent Crime." *Journal of Studies on Alcohol* 62 (5): 628–636.

Gorman, D. M., W. R. Ponicki, Q. Zheng, D. Han, P. J. Gruenewald, and A. J. Gaidus. 2018. "Violent Crime Redistribution in a City Following a Substantial Increase in the Number of Off-Sale Alcohol Outlets: A Bayesian Analysis." *Drug and Alcohol Review* 37 (3): 348–355. doi:10.1111/dar.12636.

Groff, E. R., and B. Lockwood. 2014. "Criminogenic Facilities and Crime across Street Segments in Philadelphia: Uncovering Evidence about the Spatial Extent of Facility Influence." *Journal of Research in Crime and Delinquency* 51 (3): 277–314. doi:10.1177/0022427813512494.

Gruenewald, P. J., B. Freisthler, L. Remer, E. A. LaScala, and A. Treno. 2006. "Ecological Models of Alcohol Outlets and Violent Assaults: Crime Potentials and Geospatial Analysis." *Addiction* 101 (5): 666–677. doi:10.1111/j.1360-0443.2006.01405.x.

Gruenewald, P. J., and L. Remer. 2006. "Changes in Outlet Densities Affect Violence Rates." *Alcoholism: Clinical and Experimental Research* 30 (7): 1184–1193. doi:10.1111/acer.2006.30.issue-7.

Haberman, C. P., E. R. Groff, and R. B. Taylor. 2013. "The Variable Impacts of Public Housing Community Proximity on Nearby Street Robberies." *Journal of Research in Crime and Delinquency* 50 (2): 163–188. doi:10.1177/0022427811426335.

Heffner, J. 2013. Statistics of the RTMDx utility. In*Risk terrain modeling diagnostics utility user manual, Version1.0*, edited by J. M. Caplan, L. W. Kennedy, and E. L. Piza . Newark, NJ: Rutgers Center on Public Security

Irvin-Erickson, Y. (2014). "Identifying risky places for crime: An analysis of the criminogenic spatiotemporal influences of landscape features on street robberies." Doctoral dissertation, Rutgers University-Graduate School-Newark.

Kennedy, L. W., J. M. Caplan, and E. L. Piza. 2018. *Risk-Based Policing: Evidence-Based Crime Prevention with Big Data and Spatial Analytics*. Oakland, CA: University of California Press.

Kennedy, L. W., J. M. Caplan, E. L. Piza, and H. Buccine-Schraeder. 2016. "Vulnerability and Exposure to Crime: Applying Risk Terrain Modeling to the Study of Assault in Chicago." *Applied Spatial Analysis and Policy* 9 (4): 529–548. doi:10.1007/s12061-015-9165-z.

Lipton, R., and P. Gruenewald. 2002. "The Spatial Dynamics of Violence and Alcohol Outlets." *Journal of Studies on Alcohol* 63 (2): 187–195.

Livingston, M. 2008a. "Alcohol Outlet Density and Assault: A Spatial Analysis." *Addiction* 103 (4): 619–628. doi:10.1111/j.1360-0443.2008.02136.x.

Livingston, M. 2008b. "A Longitudinal Analysis of Alcohol Outlet Density and Assault." *Alcoholism: Clinical and Experimental Research* 32 (6): 1074–1079. doi:10.1111/j.1530-0277.2008.00669.x.

Livingston, M. 2011a. "Alcohol Outlet Density and Harm: Comparing the Impacts on Violence and Chronic Harms." *Drug and Alcohol Review* 30 (5): 515–523. doi:10.1111/dar.2011.30.issue-5.

Livingston, M. 2011b. "A Longitudinal Analysis of Alcohol Outlet Density and Domestic Violence." *Addiction* 106 (5): 919–925. doi:10.1111/add.2011.106.issue-5.

Livingston, M., T. Chikritzhs, and R. Room. 2007. "Changing the Density of Alcohol Outlets to Reduce Alcohol-Related Problems." *Drug and Alcohol Review* 26 (5): 557–566. doi:10.1080/09595230701499191.

Macintyre, S., and R. Homel. 1997. "Danger on the Dance Floor: A Study of Interior Design, Crowding and Aggression in Nightclubs." *Policing for Prevention: Reducing Crime, Public Intoxication and Injury* 7: 91–113.

Madensen, T. D., and J. E. Eck. 2008. "Violence in Bars: Exploring the Impact of Place Manager Decision-Making." *Crime Prevention and Community Safety* 10 (2): 111–125. doi:10.1057/cpcs.2008.2.

Nielsen, A. L., T. D. Hill, M. T. French, and M. N. Hernandez. 2010. "Racial/Ethnic Composition, Social Disorganization, and Offsite Alcohol Availability in San Diego County, California." *Social Science Research* 39 (1): 165–175. doi:10.1016/j.ssresearch.2009.04.006.

Norström, T. 2000. "Outlet Density and Criminal Violence in Norway, 1960–1995." *Journal of Studies on Alcohol* 61 (6): 907–911.

Osgood, D. W., and J. M. Chambers. 2000. "Social disorganization outside the metropolis: An analysis of rural youth violence." *Criminology* 38 (1): 81–116.

Piza, E., S. Feng, L. Kennedy, and J. Caplan. 2017. "Place-Based Correlates of Motor Vehicle Theft and Recovery: Measuring Spatial Influence across Neighbourhood Context." *Urban Studies* 54 (13): 2998–3021. doi:10.1177/0042098016664299.

Pridemore, W. A. 2011. "Poverty Matters: A Reassessment of the Inequality-Homicide Relationship in Crossnational Studies." *British Journal of Criminology* 51 (5): 739–772. doi:10.1093/bjc/azr019.

Pridemore, W. A., and T. H. Grubesic. 2012. "Community Organization Moderates the Effect of Alcohol Outlet Density on Violence." *The British Journal of Sociology* 63 (4): 680–703. doi:10.1111/j.1468-4446.2012.01432.x.

Pridemore, W. A., and T. H. Grubesic. 2013. "Alcohol Outlets and Community Levels of Interpersonal Violence: Spatial Density, Outlet Type, and Seriousness of Assault." *Journal of Research in Crime and Delinquency* 50 (1): 132–159. doi:10.1177/0022427810397952.

Ratcliffe, J. H. 2012. "The Spatial Extent Of Criminogenic Places: a Changepoint Regression Of Violence around Bars." *Geographical Analysis* 44 (4): 302-320. doi:10.1111/j.1538-4632.2012.00856.x.

Roman, C. G., S. E. Reid, A. S. Bhati, and B. Tereshchenko. 2008. *Alcohol Outlets as Attractors of Violence and Disorder: A Closer Look at the Neighborhood Environment*. Washington, DC: Urban Institute.

Roncek, D. W., and P. A. Maier. 1991. "Bars, Blocks, and Crimes Revisited: Linking the Theory of Routine Activities to the Empiricism of "Hot Spots"." *Criminology* 29 (4): 725–753. doi:10.1111/crim.1991.29.issue-4.

Sampson, R. J., and S. W. Raudenbush. 1999. "Systematic Social Observation of Public Spaces: A New Look at Disorder in Urban Neighborhoods." *American Journal of Sociology* 105 (3): 603–651. doi:10.1086/210356.

Scribner, R., D. Cohen, S. Kaplan, and S. H. Allen. 1999. "Alcohol Availability and Homicide in New Orleans: Conceptual Considerations for Small Area Analysis of the Effect of Alcohol Outlet Density." *Journal of Studies on Alcohol* 60 (3): 310–316.

Scribner, R. A., D. P. MacKinnon, and J. H. Dwyer. 1994. "Alcohol Outlet Density and Motor Vehicle Crashes in Los Angeles County Cities." *Journal of Studies on Alcohol* 55 (4): 447–453.

Scribner, R. A., D. P. MacKinnon, and J. H. Dwyer. 1995. "The Risk of Assaultive Violence and Alcohol Availability in Los Angeles County." *American Journal of Public Health* 85 (3): 335–340.

Shaw, C. R., and H. D. McKay. 1942. *Juvenile Delinquency and Urban Areas*. Chicago, IL: University of Chicago Press.

Snowden, A. J. 2016. "Alcohol Outlet Density and Intimate Partner Violence in a Nonmetropolitan College Town: Accounting for Neighborhood Characteristics and Alcohol Outlet Types." *Violence and Victims* 31 (1): 111–123. doi:10.1891/0886-6708.VV-D-13-00120.

Snowden, A. J., and W. A. Pridemore. 2013. "Alcohol and Violence in a Nonmetropolitan College Town: Alcohol Outlet Density, Outlet Type, and Assault." *Journal of Drug Issues* 43 (3): 357–373. doi:10.1177/0022042613475788.

Speer, P. W., D. M. Gorman, E. W. Labouvie, and M. J. Ontkush. 1998. "Violent Crime and Alcohol Availability: Relationships in an Urban Community." *Journal of Public Health Policy* 303–318. doi:10.2307/3343538.

State Liquor License. 2015. Definition Of New York License Classes. Retrived from https://sla.ny.gov/

United States Census Bureau. 2015. Total Population 2011-2015 American Community Survey 5-Year Estimates [Data file and annotations]. Retrieved from https://factfinder.census.gov/faces/tableservices/jsf/pages/ productview.xhtml?pid=ACS_15_5YR_B01003&prodType=table

Värnik, A., K. Kõlves, M. Väli, L. M. Tooding, and D. Wasserman. 2007. "Do Alcohol Restrictions Reduce Suicide Mortality?" *Addiction* 102 (2): 251–256. doi:10.1111/j.1360-0443.2006.01687.x.

Waller, M. W., B. J. Iritani, S. L. Christ, H. K. Clark, K. E. Moracco, C. T. Halpern, and R. L. Flewelling. 2012. "Relationships among Alcohol Outlet Density, Alcohol Use, and Intimate Partner Violence Victimization among Young Women in the United States." *Journal of Interpersonal Violence* 27 (10): 2062–2086. doi:10.1177/0886260511431435.

White, G. F., R. R. Gainey, and R. A. Triplett. 2015. "Alcohol Outlets and Neighborhood Crime: A Longitudinal Analysis." *Crime and Delinquency* 61 (6): 851–872. doi:10.1177/0011128712466386.

Wortley, R., and L. Mazerolle. 2008. "Environmental Criminology and Crime Analysis: Situating the Theory, Analytic Approach and Application." In *Environmental Criminology and Crime Analysis*, edited by R. Wortley and L. Mazerolle, 1–15. Portland, OR: Willian.

Zhu, L., D. M. Gorman, and S. Horel. 2004. "Alcohol Outlet Density and Violence: a Geospatial Analysis." *Alcohol And Alcoholism* 39 (4): 369–375. doi:10.1093/alcalc/agh062.